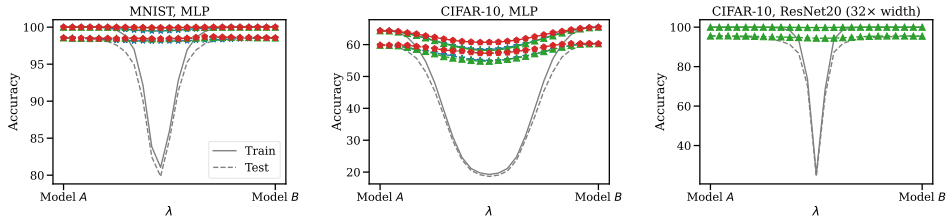

Figure 9: Top-1 accuracy results for the MNIST and CIFAR-10 models of Figure 2.

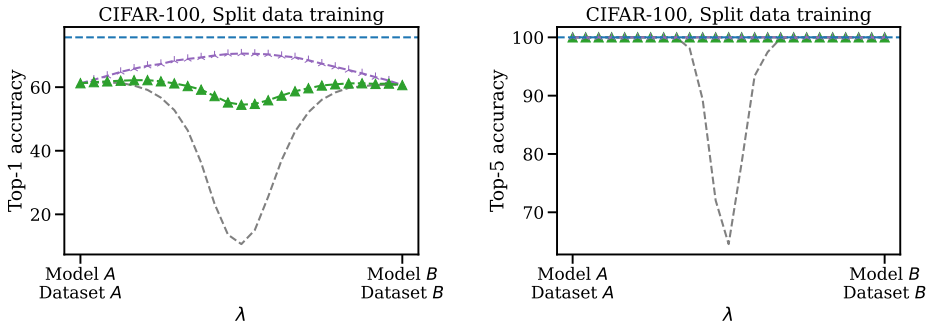

Figure 10: Accuracy results for the CIFAR-100 split data experiment.

Interpolating these matched models at $\lambda = 0.5$, we have

$$f_{\frac{1}{2}(\Theta_A + \pi_{gud}(\Theta_B))}(x) = \begin{bmatrix} 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.5 & 0 \\ 0 & (1+\epsilon)/2 \end{bmatrix} \begin{bmatrix} 1 \\ 1+\epsilon \end{bmatrix} x \qquad = (0.5 + O(\epsilon))x \tag{9}$$

$$f_{\frac{1}{2}(\Theta_A + \pi_{wm}(\Theta_B))}(x) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \epsilon/2 \end{bmatrix} \begin{bmatrix} 1+\epsilon/2 \\ 1+\epsilon/2 \end{bmatrix} x \qquad = (1+\epsilon/2)x \tag{10}$$

Here we can see that the greedy uni-directional matching (Equation (9)) results in a merged model that fails to represent the input, identity-function models. On the other hand, our weight matching algorithm (Algorithm 1, Equation (10)) produces a merged model that accurately reflects both of the input models, and even improves performance over the B model. [5]

A.8 Auxiliary Plots

A.9 Straight-through Estimator Details

See Algorithm 2 for a complete description of the straight-through estimator algorithm.

A.10 Merging Many Models

We propose Algorithm 3 to merge the weights of more than two models at a time.

Following an argument similar to Lemma 2, it can be seen that Algorithm 3 terminates.

In our limited testing, we found that this algorithm converges quickly to solutions that extrapolate better than individual models and results in a merged model with better probability estimate calibra-

[5] We note that this example can be extended to the more conventional presentation, including non-linear activation functions, by inserting ReLU activations in each layer and considering x in the positive domain $\mathbb{R}^+$. The positive domain restriction can also be lifted by instead considering the task of absolute value estimation and adding layers $\sigma \circ \begin{bmatrix} 1 & 1 \end{bmatrix} \circ \sigma \circ \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ to the beginning of the network.

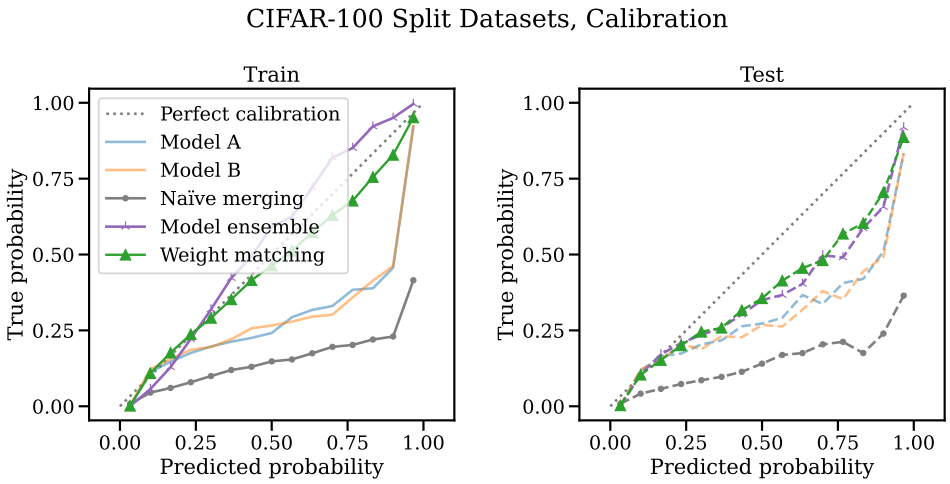

Figure 11: **Merging CIFAR-100 split data models results in superior probability calibration.** Although our merged model is not competitive in terms of top-1 accuracy in the CIFAR-100 split data experiment, we find that it has far better calibrated probability estimates than either of the input models. In addition, we achieve calibration results on par with model ensembling while requiring $2\times$ less memory and compute.

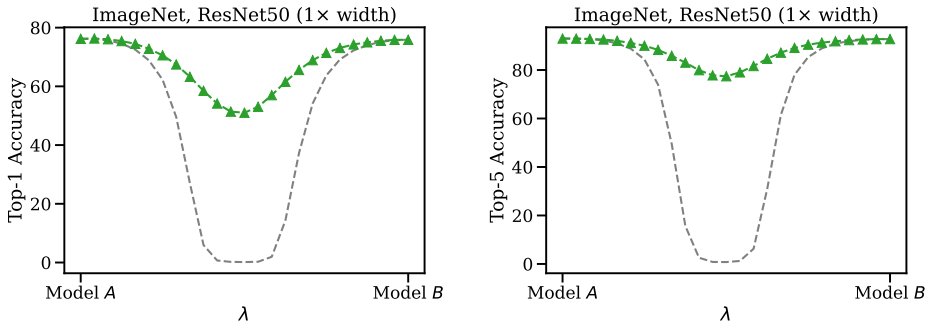

Figure 12: Accuracy results for merged ResNet50 ($1\times$ width) models on ImageNet.

Algorithm 2: Straight-through estimator training

Given: Model weights Θ_A, Θ_B, and a learning rate η.

Result: A permutation π of Θ_B such that $\mathcal{L}(\frac{1}{2}(\Theta_A + \pi(\Theta_B)))$ is approximately minimized.

Initialize: $\tilde{\Theta}_B \leftarrow \Theta_A$

repeat

- $\pi(\Theta_B) \leftarrow \text{proj}(\tilde{\Theta}_B)$ using Algorithm 1.
- Evaluate the loss of the midpoint, $\mathcal{L}(\frac{1}{2}(\Theta_A + \pi(\Theta_B)))$.
- Evaluate the gradient, $\nabla\mathcal{L}$, using $\tilde{\Theta}_B$ in place of $\pi(\Theta_B)$ in the backwards pass.
- Update parameters, $\tilde{\Theta}_B \leftarrow \tilde{\Theta}_B - \eta\nabla\mathcal{L}$.

until *convergence*

Algorithm 3: MERGEMANY

Given: Model weights $\Theta_1, \ldots, \Theta_N$

Result: A merged set of parameters $\tilde{\Theta}$.

repeat
  for $i \in$ RANDOMPERMUTATION$(1, \ldots, N)$ **do**
    $\Theta' \leftarrow \frac{1}{N-1} \sum_{j \in \{1,\ldots,N\} \setminus \{i\}} \Theta_j$
    $\pi \leftarrow$ PERMUTATIONCOORDINATEDESCENT(Θ', Θ_i)
    $\Theta_i \leftarrow \pi(\Theta_i)$
  end
until *convergence*
return $\frac{1}{N} \sum_{j=1}^{N} \Theta_j$

	Train Loss	Train Acc.	Test Loss	Test Acc.
Seed 1	0.0000	1.0000	0.1153	0.9856
Seed 2	0.0000	1.0000	0.1531	0.9854
Seed 3	0.0000	1.0000	0.1229	0.9855
Seed 4	0.0000	1.0000	0.1108	0.9865
Seed 5	0.0000	1.0000	0.1443	0.9871
MERGEMANY	0.0141	0.9952	**0.0727**	0.9831

Table 3: **Merging multiple models decreases test loss by 43%.** We train five separate MLPs on MNIST. Using Algorithm 3 we merge all these models together simultaneously. This produces a model that appears to have better out-of-distribution performance than any of the input models, with superior test loss performance. We are excited by potential applications of this methodology in federated learning and ensembling, esp. along the lines of "model soups" (Wortsman et al., 2022a).

tion than any of the input models. For example, we present the results of this algorithm on MLPs trained on MNIST in Table A.10.

In addition, we found that merging multiple models helps to calibrate the resulting model predictions. We present this effect in Figure 13.

A.11 FAILED IDEA: A METHOD FOR STEEPEST DESCENT

Imagine standing in weight space at Θ_A and trying to decide in which immediate direction to move in order to approach a Θ_B-equivalent point. There are many, many possible permutations of Θ_B – call them $\pi^{(1)}(\Theta_B), \pi^{(2)}(\Theta_B), \ldots$ – to aim for in the distance. Assuming that the loss landscape is in fact convex modulo these permutation symmetries, a natural choice would be to pick the $\pi^{(i)}(\Theta_B)$ that corresponds to the direction of steepest descent starting from Θ_A since we expect $\pi^{(i)}(\Theta_B)$ to lie in the same basin as Θ_A. In other words,

$$\min_{\pi} \left. \frac{d\,\mathcal{L}(\Theta_A + \lambda(\pi(\Theta_B) - \Theta_A))}{d\,\lambda} \right|_{\lambda=0} = \min_{\pi} \nabla\mathcal{L}(\Theta_A)^{\top}(\pi(\Theta_B) - \Theta_A) \tag{11}$$

$$= -\nabla\mathcal{L}(\Theta_A)^{\top}\Theta_A + \min_{\pi} \nabla\mathcal{L}(\Theta_A)^{\top}\pi(\Theta_B) \tag{12}$$

Now, we are tenuously in a favorable situation: $\nabla\mathcal{L}(\Theta_A)$ is straightforward to compute, and picking the best π reduces to a matching problem. In particular it is a SOBLAP matching problem of the same form as in Section 3.2. In addition, there is a fast, exact solution for the single intermediate layer case ($L = 2$).

In practice, we found that this method can certainly find directions of steepest descent, but that they are accompanied by high barriers in between the initial dip and $\pi(\Theta_B)$.